EART11300 Semester 2 Microscope Practical Class 2

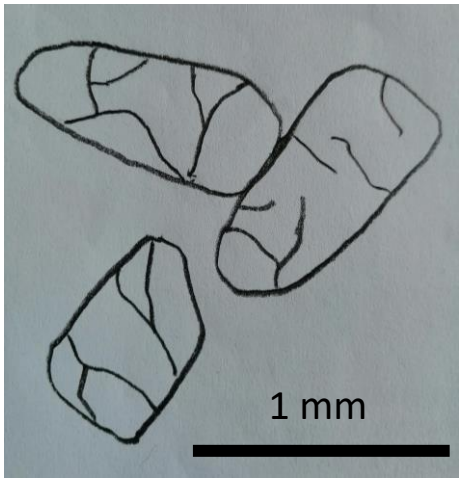
Igneous, sedimentary and metamorphic rocks under the microscope - answers

Peridotite (RP or R56)

This is an igneous rock. Most of the Earth's mantle is made of peridotite, but these samples actually comes from a large intrusion (from the island of Rum, Inner Hebrides, Scotland). You looked at R56 last week; RP is similar. The main mineral is **olivine**. It displays quite respectable crystal shapes and well-developed irregular fractures. Draw a few crystals that show a reasonable crytal shape (they will have parallel sides). Is its relief high, or low? High Can you see any cleavage in it? No (maybe weak). Note the range of interference colours represented by the various crystal orientations. What is the **maximum** colour (on the Michel Levy colour chart) that you can see? It will be a colour in the intermediate range.

Now look at the evidence that this is an igneous rock, using the possible attributes of igneous rocks:

- Does it have interlocking crystals? No.
- Do some crystals have their own shape? Yes (not so obvious in R56).
- Does it contain relatively few minerals? Yes, most of the rock is made of olivine plus two other minerals (there are also a few opaque grains and some alteration).
- Are crystals of each mineral quite uniform? Yes.
- Does it contain fossil gas bubbles (= vesicles)? No, it has no porosity.
- Do the crystals have a preferred orientation? No.



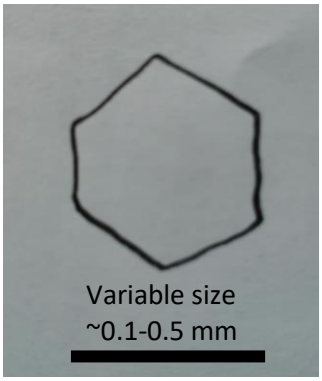
Note these crystals are not as parallel sided as they could be!

Blueschist (A/986)

This is a metamorphic rock. Before metamorphism it was an igneous rock (probably volcanic), and it has been metamorphosed at high pressure, producing several new minerals. The most distinctive new mineral is the blue one (which you do **not** need to be able to identify). Look at the **pleochroism** of this mineral. What pleochroic colours can you see? Mostly pale blue and mauve, but some crystals may go almost colourless. There is a mineral in this thin section that is equant (see 2-D sections through 3-D crystals on page 6 of last week's practical) and isotropic. Look for it and draw a sketch of a crystal. What is its relief? high. What is its colour? Colourless or very pale pink. This mineral is garnet.

Now look at the evidence that this is a metamorphic rock, using the possible attributes of metamorphic rocks:

- Does it have compositional layering? Yes. Look at the slide with the naked eye to see this.
- Does it contain relatively few minerals? Yes, 5 or 6.
- Are crystals of each mineral quite uniform? Yes.
- Do some crystals have their own shape? Yes.
- Does it contain any porosity? No.
- Do the crystals have a preferred orientation? Yes, it has foliation.



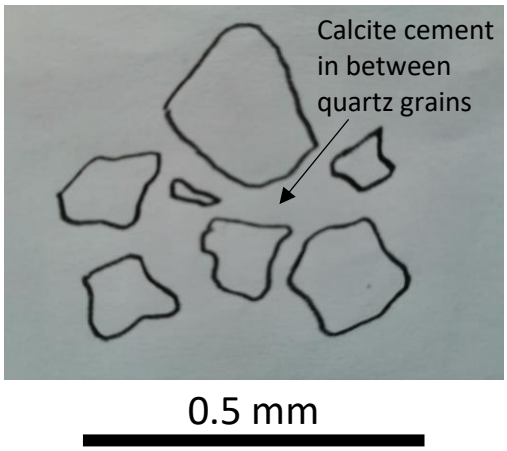
Sandstone (R25 or SK)

This is a sedimentary rock. It contains 2 minerals, one with low interference colour, one with high interference colour. Look at the low interference colour mineral. Does it contain any cleavage? No. This mineral is quartz. Notice how the crystals are all very clear – there is no alteration. This is a useful property of quartz. Other properties of quartz are: low relief, no twinning or zoning (which you will see in some low-interference colour minerals in other thin sections).

The other mineral, with high interference colour, is calcite. The sandy interference colour, with green and pink tinges, is a typial high interference colour. Calcite (and its fellow carbonate mineral, dolomite) is the only common mineral you will see with high interference colour. Notice how the calcite is filling in the gaps around the quartz crystals, forming a cement. Sketch a few quartz grains and annotate your sketch to show the calcite cement.

Now look at the evidence that this is a sedimentary rock, using the possible attributes of sedimentary rocks:

- Does it consist of particles (clasts) of pre-existing rocks? Yes, many of the quartz grains have a very irregular shape, indicating that they are not fresh crystals but have been abraded from a pre-existing rock.
- Are the clasts rounded? Some of them.
- Does it contain fossils? No.
- Does it contain organic matter? No.
- Does it contain any porosity? No (the calcite cement holding the quartz grains together has filled any earlier porosity).
- Is it layered? Not that you can see on the thin section scale.



SB2

Before working out what rock type this is, look at the green-brown mineral. It has very good cleavage – make sure you can see this. Draw a crystal showing good cleavage. Now rotate the stage so that the cleavage in this crystal is oriented N-S. Which of the following pleochroic colours can you see: pale brown or dark green-brown? Now rotate the stage so that the cleavage in this crystal is oriented E-W. Which of the following pleochroic colours can you see: pale brown or dark green-brown? This mineral is biotite (usually more of an orange-brown than a green-brown). Notice that is full of tiny inclusions with pleochroic haloes. These inclusions are the mineral zircon, which contain small amounts of radioactive uranium that emit alpha radiation that causes damage to the biotite. These are quite common in biotite.

Use the attributes of rocks in thin section to determine what rock type this is, giving reasons for your answer. Metamorphic. The crystals are aligned in a foliation.

NH

Before working out what rock type this is, look at the crystals of the colourless mineral in between the green crystals (that you looked at last week). Is their interference colour low, intermediate or high? Low. Notice how they are twinned (see description below). This mineral is feldspar (plagioclase feldspar, to be precise, but we are just going to call it feldspar today).

Use the attributes of rocks in thin section to determine what rock type this is, and draw a sketch of a few crystals to illustrate this. Metamorphic. The crystals are aligned in a foliation.

CP

This rock contains 3 main minerals plus an opaque phase and some alteration minerals. These 3 minerals are:

1. A pale brown mineral. Is this mineral pleochroic? Not obviously. Does it have cleavage? Yes (but this is not obvious in all thin sections). Is its interference colour low, intermediate or high? Intermediate (if it is pink or higher, and very bright, then it is intermediate).
2. A colourless mineral with high relief. Does this mineral have cleavage? No. Fractures? Yes Is its interference colour low, intermediate or high? Intermediate. Notice the alteration in the fractures – this is common for this mineral, which is olivine.
3. A colourless mineral with low relief. Is its interference colour low, intermediate or high? Low. Notice how this mineral is twinned (see description above). It is feldspar (plagioclase feldspar, to be precise, but we are just going to call it feldspar today).

Use the attributes of rocks in thin section to determine what rock type this is, and draw a sketch of a few crystals to illustrate this.

Igneous. Some of the crystals are interlocking and they are all in random orientation.

CG

The main minerals in this rock are:

1. **quartz** – see the R25/SK description
2. **feldspar** – see the NH and CP descriptions. In this rock the feldspar is heavily altered (to flecks that are either grey under crossed polars or have bright interference colours). This is common in feldspar, whereas quartz is never altered.
3. **mica**. Most of the mica is **biotite**. You have already seen it in SB2, but the biotite in this rock is a more typical red-brown colour. As for SB2, draw a biotite crystal showing good cleavage, and describe the pleochroic colour when the cleavage is oriented N-S, and when it is oriented E-W. Note that you also looked at biotite last week in 15216.

Use the attributes of rocks in thin section to determine what rock type this is, and explain your answer.

Igneous. Some of the crystals are interlocking and they are all in random orientation.

S10

The most abundant mineral in this rock occurs in a wide range of grain sizes. It is the same mineral as in at least one of the samples above. Can you identify it from its optical properties? Quartz: colourless, low relief, low interference colour, no twinning or alteration.

Use the attributes of rocks in thin section to determine what rock type this is, and draw a sketch to illustrate your answer.

Sedimentary. Consists of clasts of pre-existing rocks, including polycrystalline rock fragments. These clasts are rounded.

RE114

This sample contains crystals of one mineral that are up to ~5 mm long. The rest of the rock is very fine-grained. The large crystals are the same mineral as in at least one of the samples above. Can you use its optical properties to identify it?

Use the attributes of rocks in thin section to determine what rock type this is, and draw a sketch to illustrate this.

Igneous. Contains large olivine crystals. These have their own shape and are in random orientation. The rock also contains some vesicles (these are now filled with the glue that holds the thin section together; this glue is amorphous and is therefore isotropic).